

A Machine-Verified Disproof of TLMC Conjecture #458: The Möbius Number of the Tamari Lattice

Batch 8 solution submission

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Abstract

We disprove a conjecture of the Last Math Competition asserting that the absolute value of the Möbius number of the Tamari lattice T_n is $(n-2)(-1)^n$, “a closed formula verifiable up to $n = 8$ ”. In fact the Tamari Möbius numbers are $\mu(T_2) = -1$, $\mu(T_3) = 1$ (the pentagon), $\mu(T_4) = -1$; hence $|\mu(T_4)| = 1 \neq 2 = n-2$ (and already $|\mu(T_2)| = 1 \neq 0$). The formalization constructs the Tamari lattices T_2, T_3, T_4 from scratch (binary trees, right-rotation order), validates the construction against Chapoton’s interval-count theorem (T_3 has 13 and T_4 has 68 intervals), and computes the Möbius numbers by dynamic programming of the textbook recursion $\mu(x, y) = -\sum_{x \leq z < y} \mu(x, z)$ along a linear extension. All computations are kernel-verified; the main results depend on at most `[propext]`.

1 The conjecture

Conjecture 1.1 (TLMC-00000000458). The absolute value of the Möbius number of the Tamari lattice T_n is $(n-2)(-1)^n$ (a closed formula verifiable up to $n = 8$).

2 The disproof

Definition 2.1 (Tamari lattice). T_n is the lattice of binary trees with n internal nodes, ordered by the reachability of right rotations $((a, b), c) \mapsto (a, (b, c))$. The minimum is the left comb and the maximum the right comb.

Theorem 2.2. *Conjecture 1.1 is false:*

$$\mu(T_2) = -1, \quad \mu(T_3) = 1, \quad \mu(T_4) = -1,$$

so $|\mu(T_4)| = 1 \neq 2 = n-2$, and also $|\mu(T_2)| = 1 \neq 0$.

Proof. The Möbius function is computed by the textbook recursion $\mu(x, x) = 1$, $\mu(x, y) = -\sum_{x \leq z < y} \mu(x, z)$, evaluated by dynamic programming along a linear extension of T_n (the trees sorted by decreasing closure size). For T_3 (the pentagon) this gives the classical value $\mu = 1$: the two atoms contribute -1 each and the coatom between them 0. For T_4 (14 elements) the recursion yields -1 , not ± 2 . $\square$

Remark (Validation of the construction). The poset built here is the genuine Tamari lattice: the numbers of intervals of T_3 and T_4 come out as 13 and 68, matching Chapoton’s theorem $\# \text{intervals} = \frac{2}{n(n+1)} \binom{4n+1}{n-1}$ (T_5 gives 399 in the accompanying computation). The min/max comb bounds and the linear-extension property are also verified.

Remark. The “verified up to $n = 8$ ” claim of the conjecture cannot refer to the standard Tamari lattice: its Möbius numbers do not grow linearly. The formula $(n - 2)$ agrees with the truth only at $n = 3$ (the pentagon, $|\mu| = 1$).

3 Machine verification

`Batch8.lean` (namespace `TLMCBatch8`) contains: `trees/treesA` (fuel-structural enumeration of binary trees, with `trees_stable`), `coversOf` (right rotations via a rewrite system), `closureN/leT` (reachability, with `closure_stable`), `interval_counts` (Chapoton validation), `comb_bounds`, `extOf/ext_linear`, the Möbius DP `muRow/muTop`, and the refutation `tlm458_false` with the packaged data `tlm458`. Compilation via `lake env lean Batch8.lean` exits 0 with zero errors and warnings; `#print axioms` shows `trees_cards`, `trees_stable`, `tlm458_false` depend on *no* axioms, and the remaining results on at most `[propext]`. No `sorry` appears.